





CIA2C14 Data Science Essentials



Data Science Life Cycle:

CRISP-DM Framework

What is Science?

A systematically organized body of knowledge

- That tells you how to solve a problem?
- That provides you a methodology or some kind of framework to follow?

Why the need for a framework

- Focus your time
- Ensure rigor and completeness
- Enable better transition to members of the cross-functional analytic teams
 - Repeatable
 - Scale to additional analysts
 - Support validity of findings

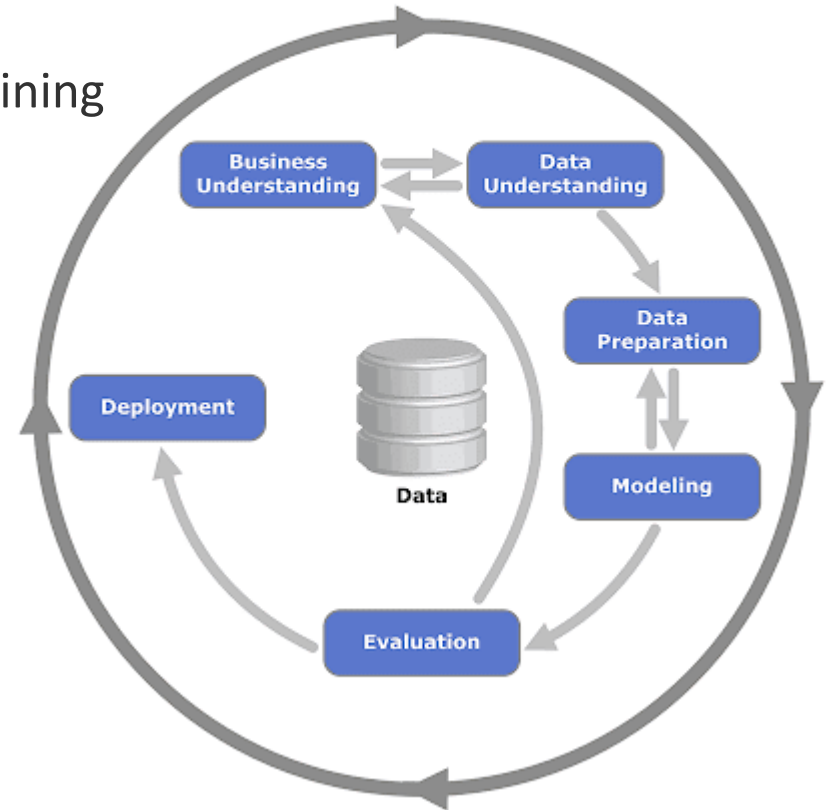
A standard process to follow

CRISP-DM

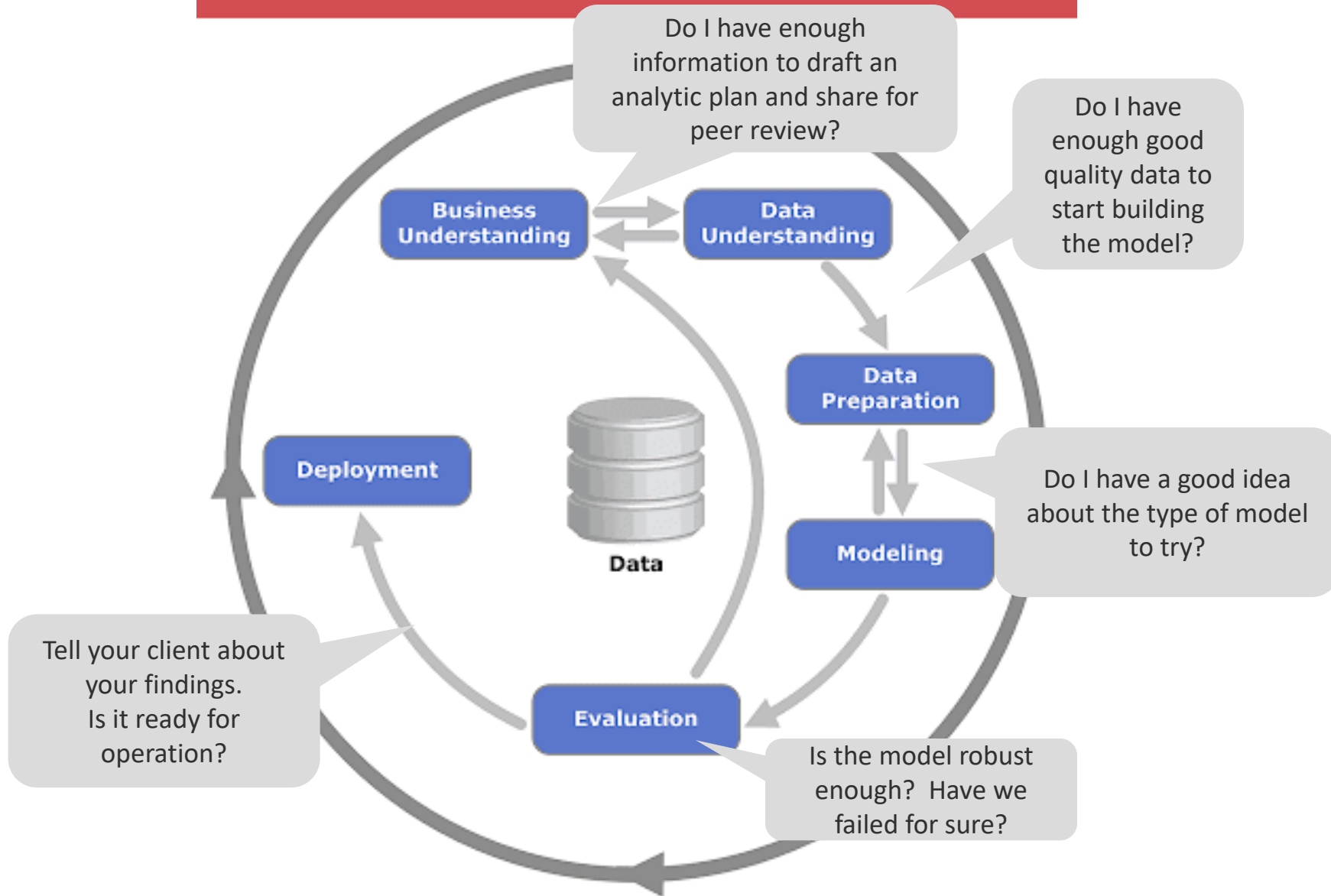
CRISP-DM:

Cross-Industry Standard Process for Data Mining

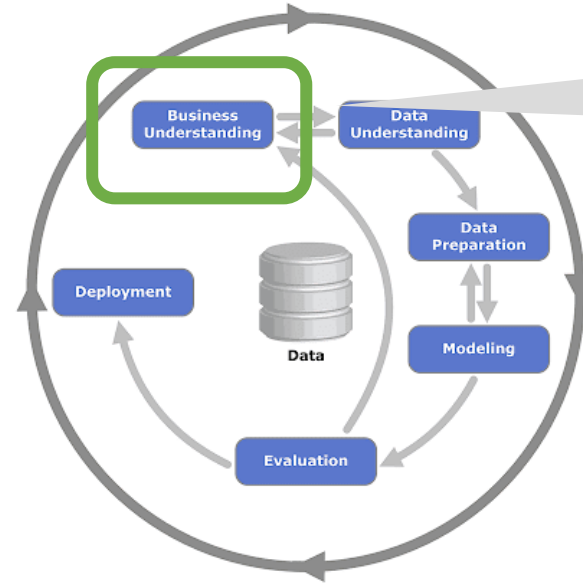
- As a **methodology**- includes descriptions of the typical phases of a project, the tasks involved with each phase, and an explanation of the relationships between these tasks.
- As a **process model**- provides an overview of the life cycle.



CRISP-DM



CRISP-DM



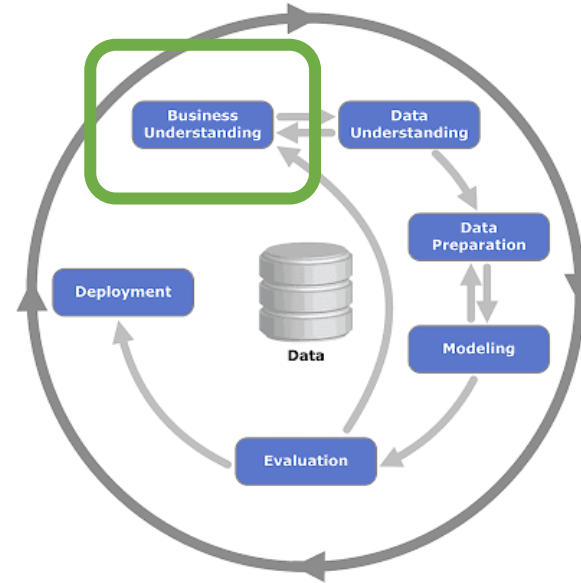
- **Learn the Business Domain**

- ▶ Determine amount of domain knowledge needed to orient you to the data and interpret results downstream
- ▶ Determine the problem type (such as clustering, classification)

- **Learn from the past**

- ▶ Have there been previous attempts in the organization to solve this problem?
- ▶ If so, why did they fail? Why are we trying again? How have things changed?

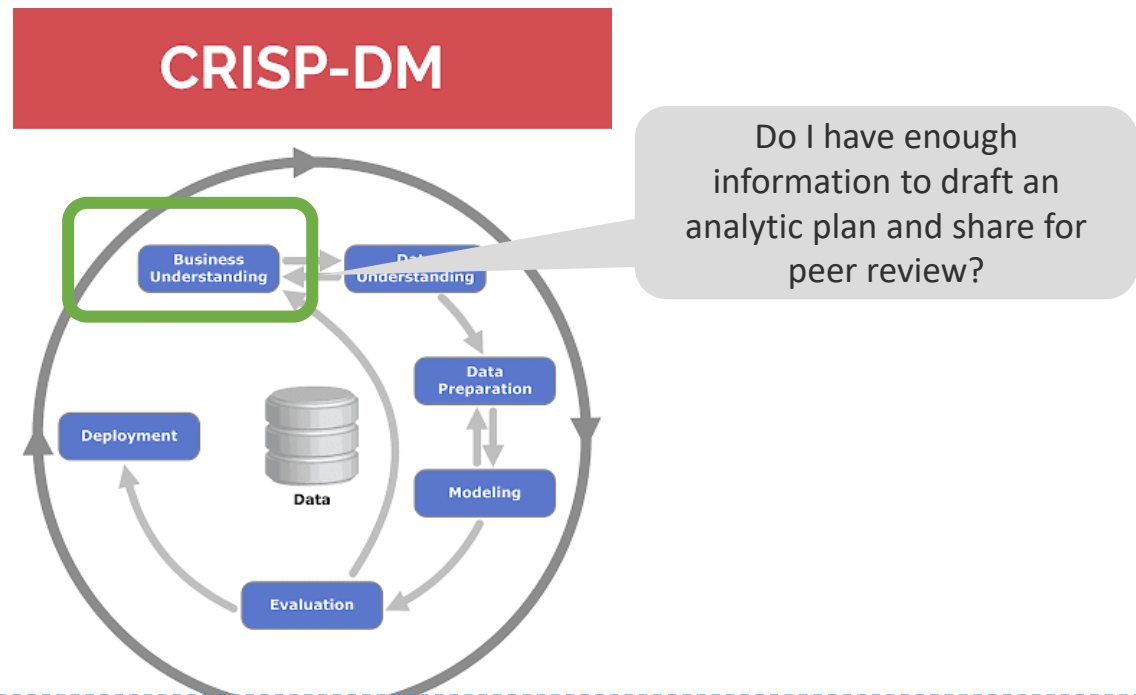
CRISP-DM



Do I have enough information to draft an analytic plan and share for peer review?

- **Resources**

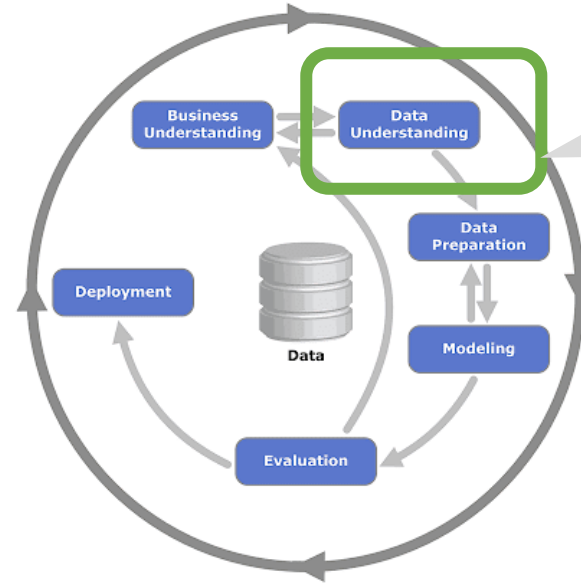
- ▶ Assess available technology
- ▶ Available data – sufficient to meet your needs
- ▶ People for the working team
- ▶ Assess scope of time for the project in calendar time and person-hours
- ▶ Do you have sufficient resources to attempt the project? If not, can you get more?



- **Frame the problem**

- ▶ *State the analytics problem*, why it is important, and to whom
- ▶ Identify key stakeholders and their interests in the project
- ▶ Clearly articulate the current situation and ***pain points***
- ▶ Objectives – identify what needs to be achieved in business terms and what needs to be done to meet the needs
 - ▶ What is the goal? What are the criteria for success? What’s “good enough”?
 - ▶ What is the failure criterion (when do we just stop trying or settle for what we have)?
- ▶ Identify the success criteria, key risks, and stakeholders (such as RACI)

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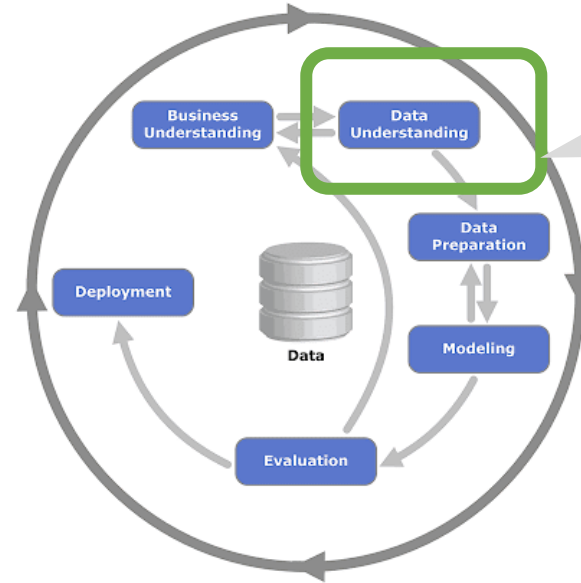


Do I have enough information to draft an analytic plan and share for peer review?

- **Formulate Initial Hypotheses**

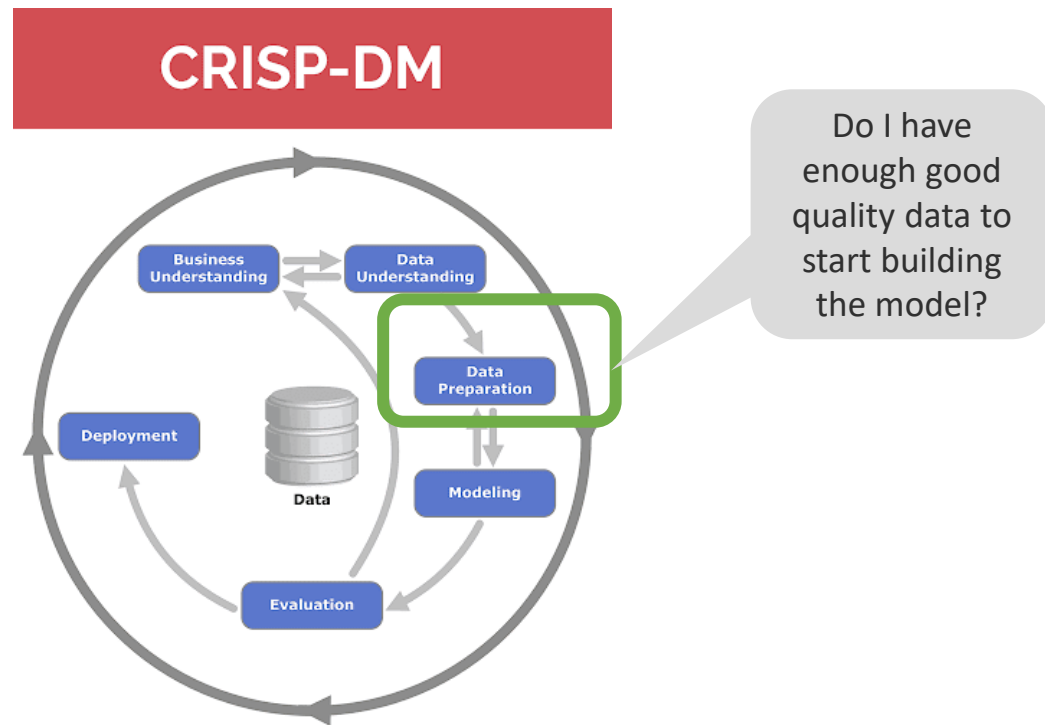
- ▶ $H_0, H_1, H_2, H_3, \dots H_n$
- ▶ Gather and assess hypotheses from stakeholders and domain experts
- ▶ Preliminary data exploration to discuss with stakeholders

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Do I have enough information to draft an analytic plan and share for peer review?

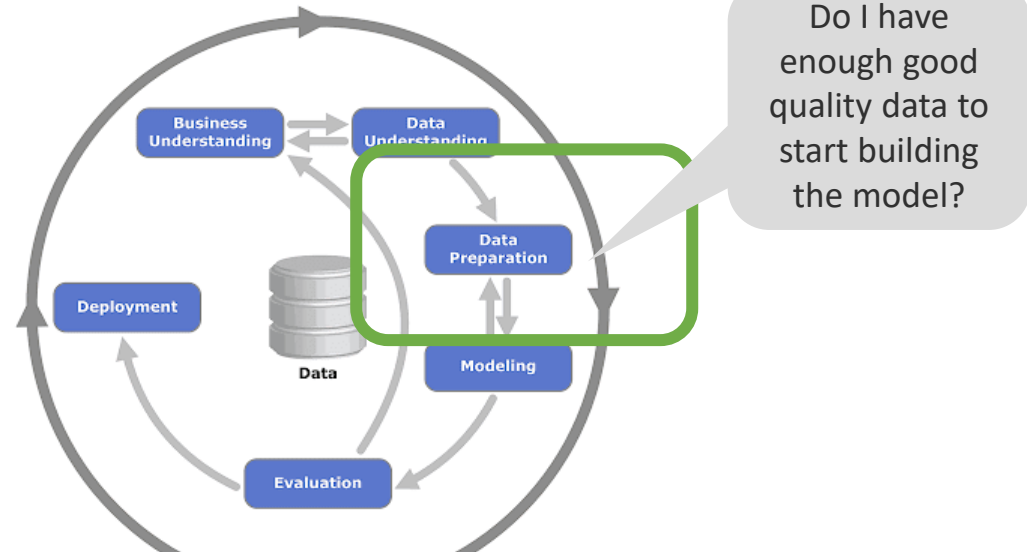
- **Identify Data Sources – Begin Learning the Data**
 - ▶ Aggregate sources for previewing the data and provide high-level understanding
 - ▶ Review the raw data
 - ▶ Determine the structures and tools needed
 - ▶ Scope the kind of data needed for this kind of problem



- **Perform ELT**

- ▶ Determine needed transformations
 - ▶ Assess data quality and structuring
 - ▶ Derive statistically useful measures
- ▶ Extract data and determine data connections for raw data, OLTP transactions, OLAP cubes or data feeds
- ▶ Big ELT and Big ETL

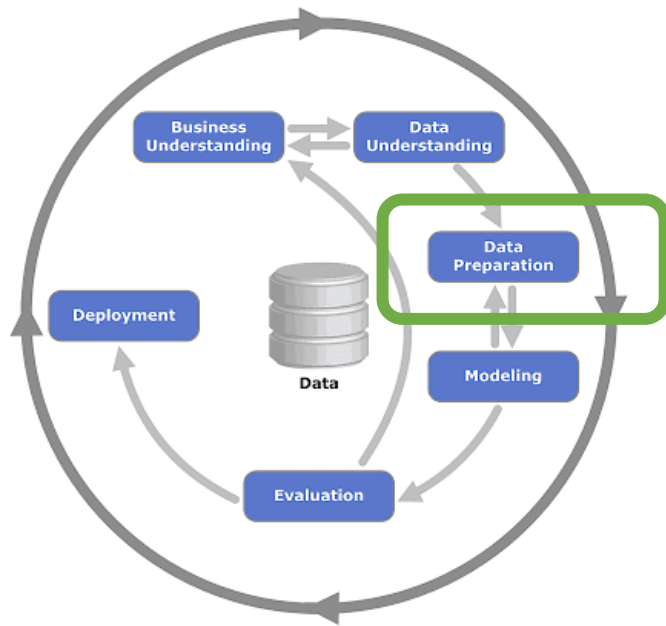
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- **Familiarize yourself with the data thoroughly**
 - ▶ List your data sources
 - ▶ What's needed vs. what's available
- **Data Conditioning**
 - ▶ Clean and normalize data
 - ▶ Discern what you keep vs. what you discard
- **Survey & Visualize**
 - ▶ Overview, zoom & filter, details-on-demand
 - ▶ Descriptive Statistics
 - ▶ Data Quality

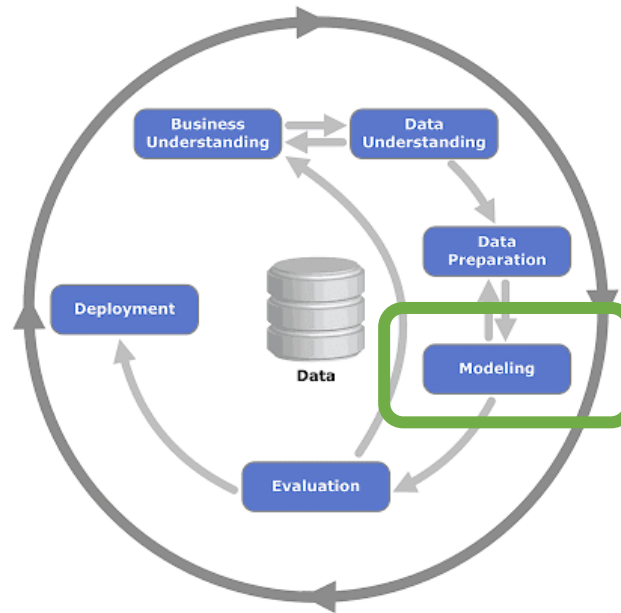
Exploratory Data Analysis

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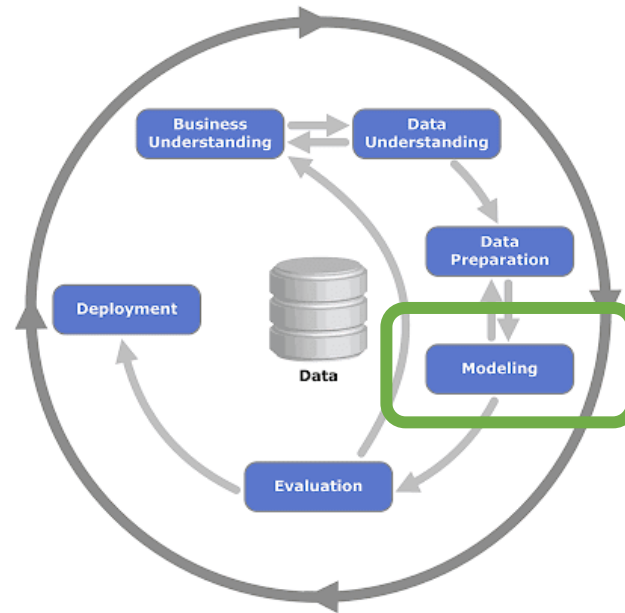
Will cover more in details in separate slides

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- **Model Planning**
 - **Determine Methods**
 - ▶ Select methods based on hypotheses, data structure and volume
 - ▶ Ensure techniques and approach will meet business objectives
 - ▶ Identify and document modeling assumptions
 - **Features Selection**
 - ▶ Inputs from stakeholders and domain experts
 - ▶ Iterative testing to confirm the most significant features

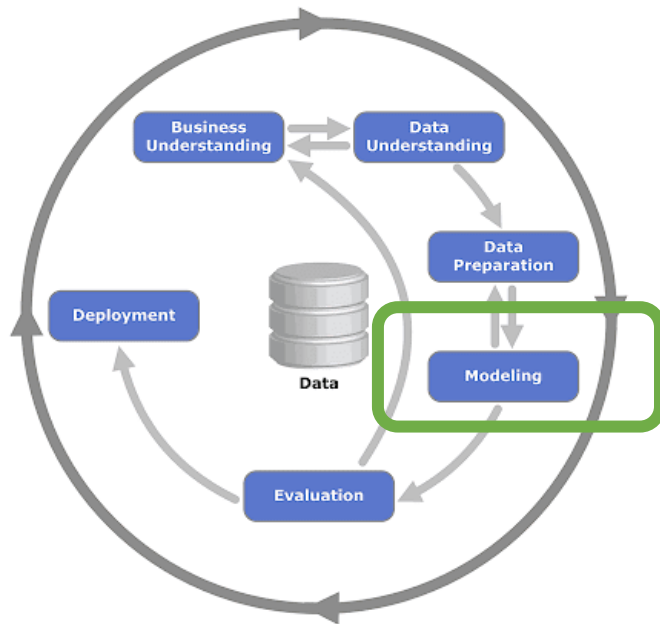
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- **Model Building**
 - **Develop data sets for training, testing and production**
 - ▶ Ensure data is sufficiently robust
 - ▶ Smaller sets for validating approach, training set for initial experiments
 - **Get the best environment you can for building models and workflows**

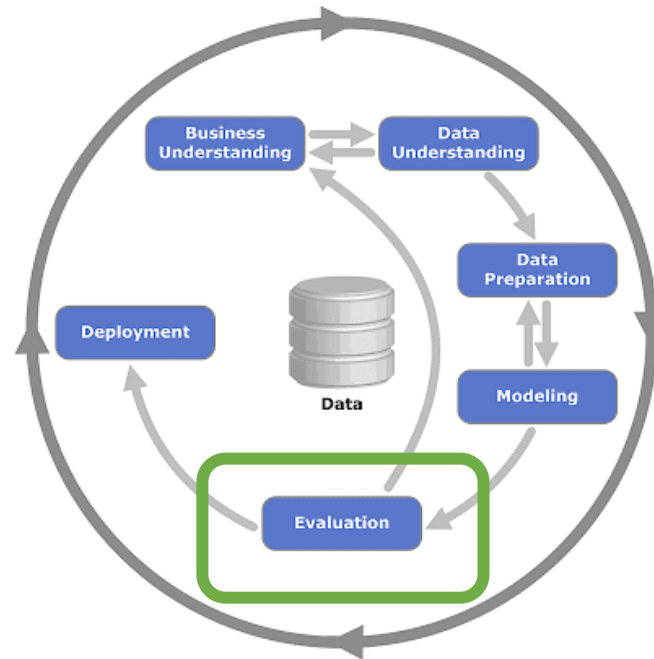
Data Modelling

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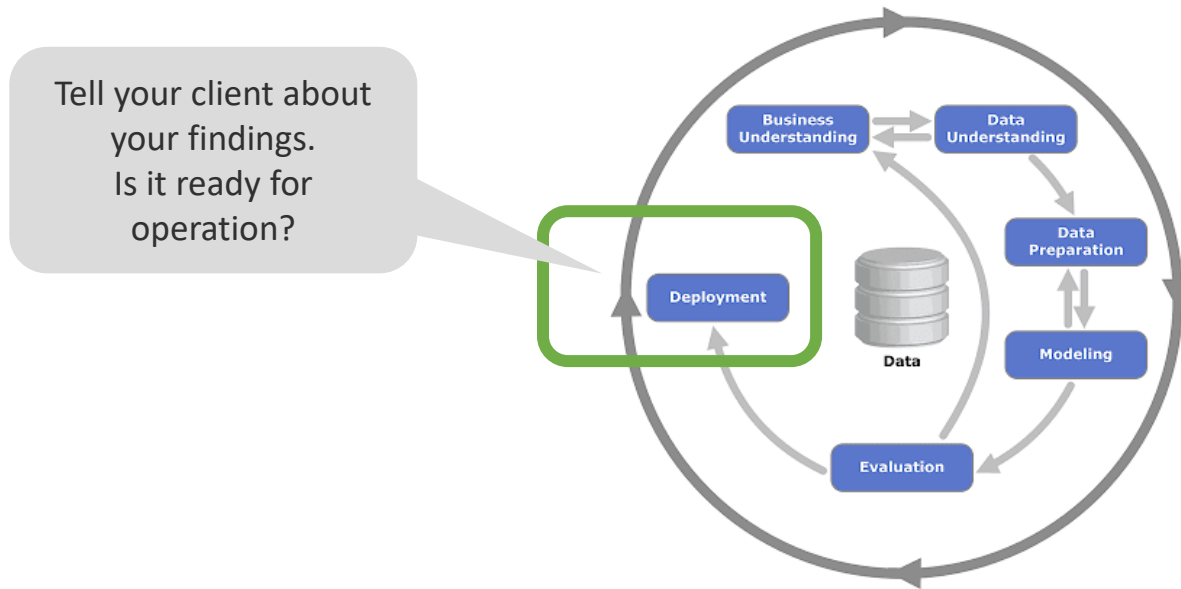
Will cover more in details in separate slides the different techniques involved

CRISP-DM



- **Communicate results**
 - *Did we succeed? Did we fail?*
 - ▶ Interpret the results
 - ▶ Compare to initial hypothesis
 - ▶ Identify key findings
 - ▶ Quantify business value
 - ▶ Summarizing findings, depending on audience

CRISP-DM



- Run a pilot
- Assess the benefits
- Deliver final deliverables
- Model Execution in Production Environment
- Define process to update and retrain the model, as needed

**WE  DATA
SCIENCE**